CSC413 Project Proposal: [Your Project Title]

Anonymous Author(s)
Affiliation
Address
email

Abstract

[Write your abstract here. Give the reader enough information about the project, data, task, and model to understand the work and determine whether they have the expertise to review it.]

1 Introduction

1.1 Problem and Motivation

[Introduce the problem and explain its real-world or research significance.]

1.2 Project Goals

[Describe the specific task, intended model behavior, and measurable objectives.]

1.3 Proposed Approach

[Briefly introduce the planned model, data, and evaluation strategy. Explain why a neural network is appropriate for the task.]

2 Background and Related Work

2.1 Background

[Explain relevant domain concepts, terminology, and task-specific challenges that readers may not already know.]

2.2 Related Work

[Discuss at least one or two prior papers that solve similar problems. For each paper, describe its task, approach, key findings, and relationship to your project.]

2.3 How Our Work Differs

[Explain what you will adapt, compare, investigate, or contribute beyond the existing work.]

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

3 Data

3.1 Dataset Description

[Describe the dataset source, size, features, labels, and data types. Discuss class balance, missing values, noise, and other relevant characteristics. Explain why the dataset is suitable for your task.]

3.2 Data Preprocessing

[Describe reproducible cleaning and formatting steps, including filtering, normalization, tokenization, resizing, encoding, or augmentation as appropriate.]

3.3 Training, Validation, and Test Splits

[Explain how the data will be split, including split proportions and how data leakage will be prevented.]

3.4 Exploratory Data Analysis

[Describe important patterns, distributions, examples, or challenges discovered during data exploration. Add plots or tables if useful.]

4 Model Architecture

4.1 Model Overview

[Describe the type of neural network you plan to use and why it fits the task and the structure of the data.]

4.2 Architecture Components

[Explain the key layers or components, their roles, and how information flows through the model. Include relevant dimensions or hyperparameters where known.]

4.3 Input and Output

[Specify the input representation, output format, prediction target, and any activation functions or loss functions you plan to use.]

4.4 Training Strategy

[Describe the planned optimizer, learning rate, batch size, number of epochs, regularization, and evaluation metrics. Mark undecided choices as preliminary plans.]

5 Model Architecture Figure

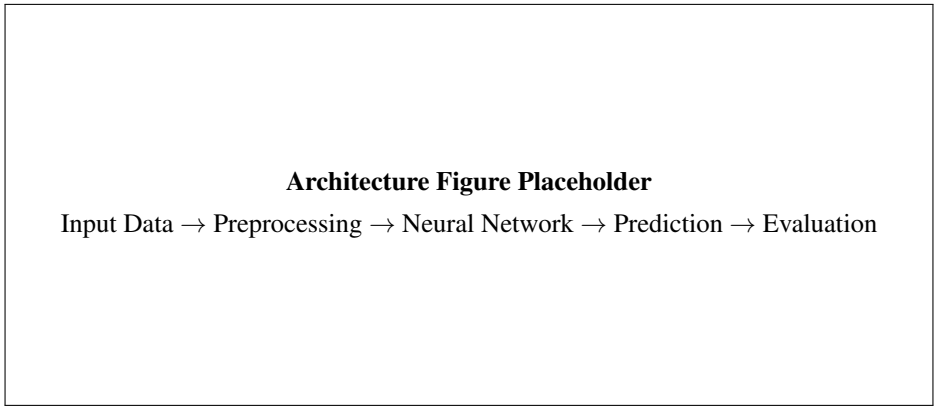


Figure 1: Overview of the proposed model architecture. Replace this placeholder with a diagram showing the actual data flow and important model components.

[Explain the key design choices illustrated in Figure **??**.]

6 Ethical Considerations

6.1 Data Collection and Privacy

[Discuss dataset consent, privacy, sensitive information, licensing, representation, and possible collection biases.]

6.2 Bias and Fairness

[Consider whether the data or model could disadvantage particular groups or systematically produce unequal errors.]

6.3 Potential Benefits and Harms

[Explain beneficial applications, misuse scenarios, limitations, and the consequences of incorrect predictions.]

6.4 Mitigation Strategies

[Describe practical steps to reduce risks, such as data filtering, privacy protection, subgroup evaluation, transparency, or human review.]

7 Work Division

Table 1: Division of responsibilities and planned time commitment.

Member	Responsibilities and Deliverables	Weeks	Hours/Week
Member One	Literature review, dataset investigation, and data preprocessing.	1–3	[X]
Member Two	Model implementation, architecture design, and training pipeline.	2–5	[X]
Member Three	Evaluation, baseline comparisons, and result visualization.	3–5	[X]
All members	Paper writing, integration, debugging, and final review.	1–6	[X]

7.1 Weekly Plan

- **Week 1:** Finalize the task, research related work, and inspect the dataset.
- **Week 2:** Complete preprocessing and establish a baseline.
- **Week 3:** Implement the proposed model and begin training.
- **Week 4:** Tune hyperparameters and run experiments.
- **Week 5:** Analyze results, perform comparisons, and address limitations.
- **Week 6:** Finalize the report, figures, and reproducibility checks.

7.2 Coordination and Dependencies

[Explain which tasks depend on others, how code and results will be integrated, how often the team will meet, and how conflicts or delays will be handled.]